NORTH BENGAL INTERNATIONAL UNIVERSITY

Lab Report : 2

TOPIC : WRITE CODE FOR ARRAY, REVERSE WAY, SUM, LARGEST NUMBER, MERGE, 5D ARRAY IN C++

COURSE CODE : CSE 0612-2122

COURSE TITLE : OBJECT ORIENTED PROGRAMMING SESSIONAL.

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CODE FOR ARRAY, REVERSE WAY, SUM, LARGEST NUMBER, MERGE, 5D ARRAY IN C++ :

Reverse way Array:

Code:

```
#include <iostream>

int main() {
    int size;
    std::cout << "Enter the size of the array: ";
    std::cin >> size;

    int arr[size];

    std::cout << "Enter the elements of the array: ";
    for (int i = 0; i < size; ++i) {
        std::cin >> arr[i];
    }

    // Reversing the array
    for (int i = 0; i < size / 2; ++i) {
        std::swap(arr[i], arr[size - 1 - i]);
    }

    std::cout << "Reversed array: ";
    for (int i = 0; i < size; ++i) {
        std::cout << arr[i] << " ";
    }
    std::cout << std::endl;

    return 0;
}
```

Output:

```
OUTPUT    TERMINAL

Enter the size of the array: 5
Enter the elements of the array: 1 2 3 4 5
Reversed array: 5 4 3 2 1
PS C:\Users\Lenovo\Documents\c++>
```

Sum and Average of Array:

Code:

```
#include <iostream>

int main() {
    int size;
    std::cout << "Enter the size of the array: ";
    std::cin >> size;

    int arr[size];
    std::cout << "Enter the elements of the array: ";
```

```
    for (int i = 0; i < size; ++i) {
        std::cin >> arr[i];
    }

    int sum = 0;
    for (int i = 0; i < size; ++i) {
        sum += arr[i];
    }

    double average = static_cast<double>(sum) / size;

    std::cout << "Sum: " << sum << std::endl;
    std::cout << "Average: " << average << std::endl;

    return 0;
}
```

Output:

```
OUTPUT    TERMINAL

Enter the size of the array: 5
Enter the elements of the array: 9 8 7 6 5
Sum: 35
Average: 7
PS C:\Users\Lenovo\Documents\c++>
```

Find the largest number from array:

Code:

```
#include <iostream>

int main() {
    int size;
    std::cout << "Enter the size of the array: ";
    std::cin >> size;

    int arr[size];
    std::cout << "Enter the elements of the array: ";
    for (int i = 0; i < size; ++i) {
        std::cin >> arr[i];
    }

    int max = arr[0];
    for (int i = 1; i < size; ++i) {
        if (arr[i] > max) {
            max = arr[i];
        }
    }

    std::cout << "The largest element in the array is: " << max <<
std::endl;

    return 0;    }
```

Output:

```
OUTPUT    TERMINAL

Enter the size of the array: 5
Enter the elements of the array: 55 45 36 13 05
The largest element in the array is: 55
PS C:\Users\Lenovo\Documents\c++>
```

Merge 2 arrays:

Code:

```cpp
#include <iostream>

int main() {
    int size1, size2;

    std::cout << "Enter the size of the first array: ";
    std::cin >> size1;
    int arr1[size1];

    std::cout << "Enter the elements of the first array: ";
    for (int i = 0; i < size1; ++i) {
        std::cin >> arr1[i];
    }

    std::cout << "Enter the size of the second array: ";
    std::cin >> size2;
    int arr2[size2];

    std::cout << "Enter the elements of the second array: ";
    for (int i = 0; i < size2; ++i) {
        std::cin >> arr2[i];
    }

    int size3 = size1 + size2;
    int arr3[size3];

    for (int i = 0; i < size1; ++i) {
        arr3[i] = arr1[i];
    }

    for (int i = 0; i < size2; ++i) {
        arr3[size1 + i] = arr2[i];
    }

    std::cout << "Merged array: ";
    for (int i = 0; i < size3; ++i) {
        std::cout << arr3[i] << " ";
    }
    std::cout << std::endl;

    return 0;
}
```

Output:

```
OUTPUT    TERMINAL

Enter the size of the first array: 4
Enter the elements of the first array: 12 23 34 45
Enter the size of the second array: 4
Enter the elements of the second array: 56 67 78 89
Merged array: 12 23 34 45 56 67 78 89
PS C:\Users\Lenovo\Documents\c++>
```

5D arrays:

Code:

```
#include <iostream>

int main() {
    const int rows = 5;
    const int cols = 5;
    const int size = rows * cols;
    int arr[size];

    // Taking input for the 1D array from the user
    std::cout << "Enter " << size << " elements for the array: ";
    for (int i = 0; i < size; ++i) {
        std::cin >> arr[i];
    }

    // Displaying the values in a 5x5 grid
    std::cout << "Array elements in a 5x5 grid:" << std::endl;
    for (int i = 0; i < rows; ++i) {
        for (int j = 0; j < cols; ++j) {
            std::cout << arr[i * cols + j] << " ";
        }
        std::cout << std::endl;
    }

    return 0;
}
```

Output:

```
OUTPUT    TERMINAL

Enter 25 elements for the array: 1 2 3 4 5 6 7 8 9 10 11 12 13 14 15 16 17 18 19 20 21 22 23 24 25
Array elements in a 5x5 grid:
1 2 3 4 5
6 7 8 9 10
11 12 13 14 15
16 17 18 19 20
21 22 23 24 25
PS C:\Users\Lenovo\Documents\c++>
```